

Ong Zheng Kai Ethan

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EDUCATION

Nanyang Technological University

Aug. 2021 – Jun. 2025

Bachelor of Engineering (Hons), Mechanical Engineering (Robotics and Mechatronics)

Singapore

- **GPA: 4.56/5.0; First Class Honours**
- Industry-Sponsored Bachelor Thesis in Collaboration with HMGICS: 3D Gaussian Splatting Techniques in Reconstructing Car Components for Automotive Assembly
- Relevant Courses: Engineering Graphics, Intro to Data Science and Artificial Intelligence, Measurements and Sensing Systems, Mechatronics Systems Interfacing, Mechatronics Engineering Design

WORK EXPERIENCE

Shopee

Jul. 2025 – Present

Project Manager

Singapore

- Managing AI Chatbot Projects across Backend, Data Science, and Product teams in China, Indonesia, and Singapore to improve customer satisfaction, automate customer service workflows, and ensure trust and safety
- Coordinating PRD reviews, technical discussions on target metrics and systems design, and conducting regular A/B testing updates to ensure process governance and stakeholder expectations are met
- Leveraged Jira, Google Sheets, and internal Project Management Tools to track development and testing progress, forecast technical and administrative blockers, and improve overall timeline and manpower efficiency

Panasonic R&D Center Singapore

Jan. 2025 – Jun. 2025

Research and Development Intern

Singapore

- Improved the realism of 3D indoor reconstruction via SLAM, point cloud editing, and radiance field techniques
- Reduced rendering times by 200% by introducing async data loading in React and Javascript, fixed vertex shader transformations, coupled [Three.js](#) mesh-splat materials, and simplified WebGL fragment shader ray tracing logic
- Dockerized multiple research papers' code for seamless deployment and testing across multiple Linux servers
- Linked MongoDB assets database to user-friendly React frontend via Axios for users to add and edit furniture and home lighting, supported Gaussian-based Phong reflections with fragment shaders written in GLSL

Mecatron

Oct. 2024 – Jun. 2025

Software Lead

Singapore

- Champions, Singapore Autonomous Underwater Vehicle Competition 2025, besting 40 teams from 20 countries
- Managed 10 SWEs via project roadmaps, documentation, code reviews, and sprints in Notion; Designed control flows with UML diagrams, and built a CI/CD pipeline in GitHub Actions and Docker to run test trees
- Created action servers and pub/sub frameworks in ROS (Python and C++), planned missions using Behavior Trees; configured Pixhawk controllers for multi-servo/relay & Ardupilot Sub setups, SITL testing in UnitySim
- Streamlined recruitment and training via an OpenCV and Python screening task, and a comprehensive tutorial to teach UnitySim with ROS TCP, Behavior Trees, and Git version control concepts specific to our workflow

Hyundai Motor Group Innovation Center Singapore

Aug. 2024 – May 2025

Student Researcher

Singapore

- Improved digital reconstruction of car parts using 3D Gaussian Splatting for vision-based motion-planning
- Optimised image dataset by 65%, replanned robotic arm trajectories via ABB RAPID and Simulator
- Evaluated image similarities via OpenCV, and point cloud ICP alignment using CloudCompare and Open3D
- Constructed Photoneo PhoXi data capture, YOLO object segmentation, and scikit-learn ML clustering pipeline

Fling Asia

Jun. 2023 – Apr. 2024

Software Engineer | Dec. 2023 – Apr. 2024

Bangkok, Thailand

- Directed marketing initiatives, product demonstrations, and inventory audits for Fling's drone operations and software across 10+ warehouses, securing ongoing stock-checking contracts with over 90% of them
- Led development of a cross-platform bulk labelling program in PySide6 and C++ in Qt, supporting Zebra printers and QR encoding, successfully deployed in Ceva Logistics Shell and DHL Harley-Davidson, Thailand

Engineering Intern | Jun. 2023 – Nov. 2023

- Designed, 3D printed, & ran FEA for drone-lights in Fusion360 for AprilTag pose alignment in dim conditions
- Improved ML inference speeds on video streams by 300% using Python multi-processing and PyTorch tensors
- Developed internal Tkinter GUI for SQL scripting, MariaDB database editing, & Google Cloud CLI commands

Republic of Singapore Air Force

Oct. 2018 – Jul. 2021

Weapon Systems Officer (Fighter) Trainee

Singapore

- Appointed as Commander Air Training Group, coordinating flight arrangements, visa processing, health certifications, manpower forecasts to overseas detachments, and disciplinary actions of 200+ aircrew candidates
- Practised crew resource management, Instrument Flight Rules, and tactical maneuvers in France and Australia
- Planned flight critical safety and survival re-training curriculums as an Operations & Life Support Systems Officer

PROJECTS AND EXTRA CURRICULAR ACTIVITIES

NTU Overseas Entrepreneurship Programme

May 2023 – Dec. 2023

- Selected from a large pool of candidates to gain entrepreneurial experience with overseas start-ups
- Provided solutions to product and business issues while working closely with start-up founder
- Attended entrepreneurship workshops and provided detailed feedback and business plans for assigned start-up

Dyson-NTU Product Development Challenge

Jan. 2023 – May 2023

Project Title: Puffin - Cigarette Smoke Filtration Device

- Created a cigarette attachment to filter toxic particles, reducing health risks associated with secondhand smoke
- Collaborated with 3 team members to conduct market research through surveys, competitor analysis, focus groups and industry trend reports, using systems design methodologies to define our product vision
- Prototyped iteratively via CAD in SolidWorks, 3D printing, soldering of power delivery boards, & PM2.5 testing
- Presented our product to Dyson engineers & directors prior to James Dyson Award submission. Final grade: A-

Espressif Systems U Amaze Venture

Mar. 2023 – Apr. 2023

First Runners-Up

- Bested 11 teams in autonomously navigating a maze via obstacle avoidance with LiDAR while performing target alignment and collision using AprilTags for pose estimation
- Fabricated an autonomous mobile robot from scratch. Hardware was created using 3D printed parts, soldering, and breadboard circuitry. Software stack includes ROS, ESP32, and PID control on a differential drive system

NTU Students' Union Ministerial Open Discussions and Events Committee

Aug. 2022 – May 2023

Events Officer

- Spearheaded events geared towards open discussion forums between Singapore Ministers and NTU students
- Directed event proposals and cash flows towards logistics, catering, manpower, and building marketing materials
- Chief Facilitator for ForwardSG Youth Talk, moderating discussions on behalf of the National Youth Council

CERTIFICATIONS, SKILLS & INTERESTS

- **Certifications:** Nvidia Deep Learning Institute: Getting Started with Deep Learning
- **Programming Languages:** C, C++, Python, GLSL, SQL, Javascript
- **Tools:** ROS, Docker, Git, Notion, Linux, OpenCV, Open3D, PyTorch, Unity, Qt, Arduino, ESP, WebGL, [Three.js](#), [Node.js](#), React, Fusion360, SolidWorks, MongoDB, Google Cloud, Microsoft Office, Photoshop
- **Languages:** Fluent in English and Mandarin, Basic conversational Thai